



Department of Computer Science and Engineering (Data Science)

Experiment 10

(Sampling Methods)

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Aim: Perform different sampling methods for creating samples on a given dataset.

Sampling is a technique of selecting individual members or a subset of the population to make statistical inferences from them and estimate characteristics of the whole population. Different sampling methods are widely used by researchers in market research so that they do not need to research the entire population to collect actionable insights.

It is also a time-convenient and a cost-effective method and hence forms the basis of any research design. Sampling techniques can be used in a research survey software for optimum derivation.

For example, if a drug manufacturer would like to research the adverse side effects of a drug on the country's population, it is almost impossible to conduct a research study that involves everyone. In this case, the researcher decides a sample of people from each demographic and then researches them, giving him/her indicative feedback on the drug's behavior.

Types of sampling: sampling methods

Sampling in market research is of two types – probability sampling and non-probability sampling. Let's take a closer look at these two methods of sampling.

1. **Probability sampling:** Probability sampling is a sampling technique where a researcher sets a selection of a few criteria and chooses members of a population randomly. All the members have an equal opportunity to be a part of the sample with this selection parameter.
2. **Non-probability sampling:** In non-probability sampling, the researcher chooses members for research at random. This sampling method is not a fixed or predefined selection process. This makes it difficult for all elements of a population to have equal opportunities to be included in a sample.

In this blog, we discuss the various probability and non-probability sampling methods that you can implement in any market research study.

Types of probability sampling with examples:



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Probability sampling is a sampling technique in which researchers choose samples from a larger population using a method based on the theory of probability. This sampling method considers every member of the population and forms samples based on a fixed process.

For example, in a population of 1000 members, every member will have a 1/1000 chance of being selected to be a part of a sample. Probability sampling eliminates bias in the population and gives all members a fair chance to be included in the sample.

- **Simple random sampling:** One of the best probability sampling techniques that helps in saving time and resources, is the Simple Random Sampling method. It is a reliable method of obtaining information where every single member of a population is chosen randomly, merely by chance. Each individual has the same probability of being chosen to be a part of a sample. For example, in an organization of 500 employees, if the HR team decides on conducting team building activities, it is highly likely that they would prefer picking chits out of a bowl. In this case, each of the 500 employees has an equal opportunity of being selected.
- **Cluster sampling:** Cluster sampling is a method where the researchers divide the entire population into sections or clusters that represent a population. Clusters are identified and included in a sample based on demographic parameters like age, sex, location, etc. This makes it very simple for a survey creator to derive effective inference from the feedback. For example, if the United States government wishes to evaluate the number of immigrants living in the Mainland US, they can divide it into clusters based on states such as California, Texas, Florida, Massachusetts, Colorado, Hawaii, etc. This way of conducting a survey will be more effective as the results will be organized into states and provide insightful immigration data.
- **Systematic sampling:** Researchers use the systematic sampling method to choose the sample members of a population at regular intervals. It requires the selection of a starting point for the sample and sample size that can be repeated at regular intervals. This type of sampling method has a predefined range, and hence this sampling technique is the least time-consuming. For example, a researcher intends to collect a systematic sample of 500 people in a population of 5000. He/she numbers each element of the population from 1-5000 and will choose every 10th individual to be a part of the sample ($\text{Total population} / \text{Sample Size} = 5000 / 500 = 10$).
- **Stratified random sampling:** Stratified random sampling is a method in which the researcher divides the population into smaller groups that don't overlap but represent



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the entire population. While sampling, these groups can be organized and then draw a sample from each group separately. For example, a researcher looking to analyze the characteristics of people belonging to different annual income divisions will create strata (groups) according to the annual family income. Eg – less than \$20,000, \$21,000 – \$30,000, \$31,000 to \$40,000, \$41,000 to \$50,000, etc. By doing this, the researcher concludes the characteristics of people belonging to different income groups. Marketers can analyze which income groups to target and which ones to eliminate to create a roadmap that would bear fruitful results.

Lab Assignments to complete in this session

1. Consider a randomly generated dataset which has id, group number (1-10) and Values as three attributes. Implement systematic sampling, stratified sampling (3 stratas) and Cluster sampling.

```
[ ] dict1={'id':random.sample(range(1,201),200),
          'group number':np.random.randint(11,size=(200)).tolist(),
          'values':np.random.randint(100,size=(200)).tolist()
        }

[ ] df=pd.DataFrame(dict1)
```

	id	group number	values
0	137	3	96
1	12	4	4
2	138	8	9
3	64	0	45
4	68	10	13
...	...	...	...
195	140	7	65
196	29	3	32
197	174	5	75
198	74	3	10
199	72	5	10

200 rows x 3 columns

```
[ ] value_class =np.arange(0,100,10)
value_class_labels = [f"{i}-{i+10}" for i in range(0,90,10)]
df['Value Class'] = pd.cut(df['values'],bins=value_class,labels=value_class_labels)
```



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Systematic Sampling

```
def systematic_sampling(df, step):  
    indexes = np.arange(0, len(df), step=step)  
    systematic_sample = df.iloc[indexes]  
    return systematic_sample  
  
[ ] systematic_sample = systematic_sampling(df, 3)  
  
display(systematic_sample)
```

	id	group number	values	Value Class
0	137	3	96	NaN
3	64	0	45	40-50
6	171	10	90	80-90
9	151	1	62	60-70
12	39	8	63	60-70
...	...	...	...	...
186	30	7	60	50-60
189	26	9	95	NaN
192	134	8	58	50-60
195	140	7	65	60-70
198	74	3	10	0-10

67 rows x 4 columns

Stratified Sampling

```
[ ] df.groupby('group number', group_keys=False).apply(lambda x: x.sample(2))
```

	id	group number	values	Value Class
117	28	0	13	10-20
11	181	0	98	NaN
84	15	1	25	20-30
9	151	1	62	60-70
138	13	2	13	10-20
28	9	2	23	20-30
107	153	3	82	80-90
158	130	3	81	80-90
77	168	4	40	30-40
118	99	4	1	0-10
76	71	5	61	60-70
55	118	5	72	70-80
122	23	6	49	40-50
27	40	6	10	0-10
152	45	7	15	10-20
87	125	7	80	70-80
171	114	8	17	10-20
183	195	8	76	70-80
162	61	9	60	50-60
46	88	9	39	30-40
79	60	10	82	80-90



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```
[ ] df.groupby('group number', group_keys=False).apply(lambda x: x.sample(frac=0.6))
```

	id	group number	values	Value Class
	20	44	0	19
	177	200	0	7
	117	28	0	13
	58	69	0	87
	11	181	0	98
	...	...	...	...
	67	63	10	45
	54	199	10	83
	24	3	10	39
	37	191	10	70
	136	95	10	44

120 rows x 4 columns

Clustered Sampling

```
clusters = np.random.choice(np.arange(1,10),size=5, replace=False)

cluster_sample = df[df['values'].isin(clusters)]

cluster_sample
```

	id	group number	values	Value Class
	1	12	4	4
	2	138	8	9
	62	101	8	9
	74	14	7	1
	104	84	2	7
	118	99	4	1
	175	79	5	9
	177	200	0	7
	191	198	5	9
	193	62	7	1

```
[ ] cluster_sample['group number'].value_counts()
```

```
4    2
8    2
7    2
5    2
2    1
0    1
Name: group number, dtype: int64
```



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- Consider titanic dataset and implement strata sampling and cluster sampling.

Strata Sampling

```
[ ] df.groupby('Sex', group_keys=False).apply(lambda x: x.sample(2))
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
345	346	1	2	Brown, Miss. Amelia "Mildred"	female	24.0	0	0	248733	13.0000	S
796	797	1	1	Leader, Dr. Alice (Farnham)	female	49.0	0	0	17465	25.9292	S
429	430	1	3	Pickard, Mr. Berk (Berk Trembisky)	male	32.0	0	0	SOTON/O.Q. 392078	8.0500	S
582	583	0	2	Downton, Mr. William James	male	54.0	0	0	28403	26.0000	S

Cluster Sampling

```
clusters = np.random.choice(np.arange(1,max(df['Age'])), size=10, replace=False)

cluster_sample = df[df['Age'].isin(clusters)]

|

cluster_sample
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
	6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625
	10	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000
	22	23	1	3	McGowan, Miss. Anna "Annie"	female	15.0	0	0	330923	8.0292
	27	28	0	1	Fortune, Mr. Charles Alexander	male	19.0	3	2	19950	263.0000
	44	45	1	3	Devaney, Miss. Margaret Delia	female	19.0	0	0	330958	7.8792
	...	...	...	...	...	...	...	...	...	...	...
	875	876	1	3	Najib, Miss. Adele Kiamie "Jane"	female	15.0	0	0	2667	7.2250
	877	878	0	3	Petroff, Mr. Nedelio	male	19.0	0	0	349212	7.8958



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```
[ ] cluster_sample['Age'].value_counts()
```

```
19.0    25
25.0    23
36.0    22
17.0    13
4.0     10
54.0     8
49.0     6
15.0     5
11.0     4
7.0      3
Name: Age, dtype: int64
```